25900319 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBCA201 - FEATURE ENGINEERING

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Why can't mathematical operations be directly applied to categorical data? [U] / [CO1]
2. Define continuous data and give a finance-related example. [K] / [CO1]
3. Define data with an example. [K] / [CO1]
4. Why is data cleaning important in feature engineering? [U] / [CO2]
5. Why does normalization improve the performance of distance-based algorithms? [U] / [CO2]
6. What is handling missing data? [K] / [CO2]
7. Write the formula to calculate bin width in Equal Width discretization. [A] / [CO3]
8. What is the purpose of using polynomial and interaction features in machine learning models? [A] / [CO3]
9. List the main steps in feature selection. [U] / [CO4]

10. Compare the practical advantages and limitations of filter, wrapper, and hybrid feature selection methods with examples. [An] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **3** questions

Each question carries **5** marks

11. Explain with examples how discrete data is different from continuous data. [U] / [CO1]

12. Define feature scaling. List and explain different methods of feature scaling with suitable formulas and examples. [K] / [CO2]

13. Discuss the importance of interaction features in real-world applications (e.g., marketing, healthcare, finance). Provide at least two examples. [A] / [CO3]

14. Compare the advantages and disadvantages of Label Encoding with examples. [An] / [CO4]

15. Explain with reasoning why PCA may reduce accuracy in classification tasks. [An] / [CO4]

[5x3 = 15]